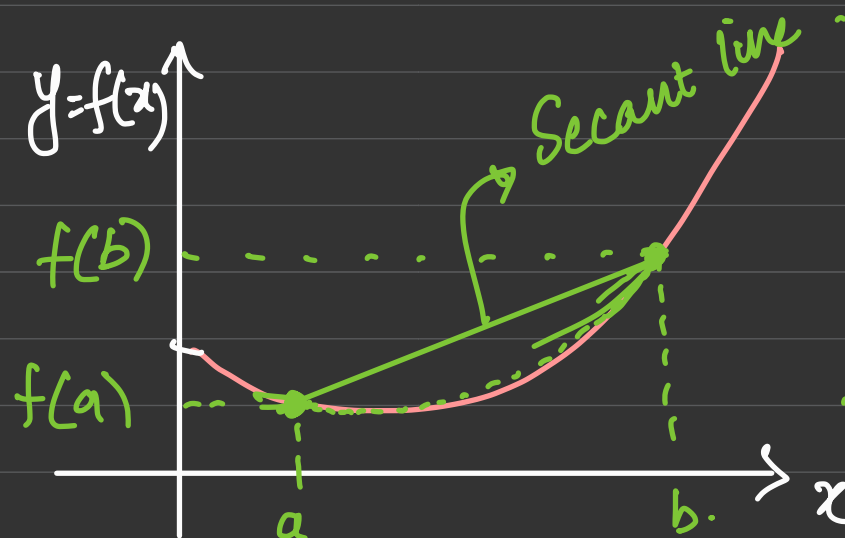


Module 4: Vector Calculus & Matrix Differentiation.

* Single Variable Calculus:



At a time $f'(t)$, the derivative of f , is defined as

$$f'(t) = \frac{df}{dt} = \lim_{\Delta t \rightarrow 0} \frac{f(t + \Delta t) - f(t)}{\Delta t}.$$

Let $f(t)$ be the distance at time t

$f(t + \Delta t) \rightarrow$ Distance at time $t + \Delta t$.

$$\Delta f = f(t + \Delta t) - f(t) = \text{Change in distance between } t \text{ \& } (t + \Delta t)$$

$$\frac{\Delta f}{\Delta t} = \frac{\text{Change in distance}}{\text{Change in time}} = \text{Average velocity.}$$

$$\frac{df}{dt} = \lim_{\Delta t \rightarrow 0} \frac{\Delta f}{\Delta t} \rightarrow \text{Derivative: Limit of average velocity.}$$

$\rightarrow$ Instantaneous speed.

Derivatives $\frac{d^n f}{dx^n}$ for $n=1, 2, \dots$ allows us to study the function f in detail.

Characteristic fn.

PDF: $f_x(x)$

CDF: $F_x(x)$

$= P(X \leq x)$

$$E[e^{i\omega x}] = \underline{\underline{\phi_x(\omega)}}$$

$$= \int_{-\infty}^{\infty} f_x(x) e^{i\omega x} dx.$$

$$E[g(x)] = \sum_i g(x_i) p_x(x_i) = \int_{-\infty}^{\infty} g(x) f_x(x) dx$$

$$\begin{aligned}\frac{d\phi_x(\omega)}{d\omega} &= \frac{d}{d\omega} \int_{-\infty}^{\infty} f(x) e^{i\omega x} dx \\ &= \int_{-\infty}^{\infty} f(x) \frac{d}{d\omega} (e^{i\omega x}) dx\end{aligned}$$

$$\frac{d\phi_x(\omega)}{d\omega} = \int_{-\infty}^{\infty} ix f(x) e^{i\omega x} dx.$$

$$\left. \frac{d\phi_x(\omega)}{d\omega} \right|_{\omega=0} = \int_{-\infty}^{\infty} ix f(x) e^{i(0)x} dx.$$

$$= \int_{-\infty}^{\infty} ix f(x) dx = i \int_{-\infty}^{\infty} x f(x) dx.$$

$$\frac{1}{i} \left. \frac{d\phi_x(\omega)}{d\omega} \right|_{\omega=0} = E[x]$$

$$\frac{d^2 \Phi_X(\omega)}{d\omega^2} = \frac{d}{d\omega} \int_{-\infty}^{\infty} ix f_X(x) e^{i\omega x} dx.$$

$$= \int_{-\infty}^{\infty} ix f_X(x) \frac{d}{d\omega} (e^{i\omega x}) dx$$

$$\frac{d^2 \Phi_X(\omega)}{d\omega^2} = \int_{-\infty}^{\infty} (ix)^2 f_X(x) e^{i\omega x} dx$$

$$\left. \frac{d^2 \Phi_X(\omega)}{d\omega^2} \right|_{\omega=0} = \int_{-\infty}^{\infty} i^2 x^2 f_X(x) dx$$

$$= i^2 \int_{-\infty}^{\infty} x^2 f_X(x) dx = (i^2) E[X^2]$$

$$\Rightarrow \frac{1}{(i)^2} \left. \frac{d^2 \Phi_X(\omega)}{d\omega^2} \right|_{\omega=0} = E[X^2]$$

$$E[x^n] = \frac{1}{i^n} \frac{d^n \phi_x(\omega)}{d\omega^n} \bigg|_{\omega=0}.$$

$E[x^2]$: Mean of Squares.
Mean Squared.

* The derivative of $cf(x)$, where c is a scalar

$$\frac{d}{dx}(cf(x)) = c \frac{d}{dx}(f(x)) = cf'(x).$$

* The derivative of $f(x) + g(x)$ is given by

$$\frac{d}{dx}(f(x) + g(x)) = \frac{df}{dx} + \frac{dg}{dx} = f'(x) + g'(x)$$

* Let $A^{m \times n}$ operate on $\vec{x}, \vec{y} \in \mathbb{R}^n$

$$A(x+y) = Ax + Ay$$

A: Linear Transformation.

$$A(c\vec{x}) = c(A\vec{x})$$

Differential is linear transformation/operator.

↳ Matrix

Sum Rule: $\frac{d}{dx}(u(x) + v(x)) = \frac{d}{dx}(u+v) = \frac{du}{dx} + \frac{dv}{dx}$

Rule of
linearity

$$\begin{aligned} \frac{d}{dx}(\alpha u(x) + \beta v(x)) &= \frac{d}{dx} \alpha(u(x)) + \frac{d}{dx} (\beta v(x)) \\ &= \alpha \frac{du}{dx} + \beta \frac{dv}{dx} \end{aligned}$$

* Product rule: $\frac{d}{dx} (u(x)v(x)) = \frac{d}{dx} (uv) = u \frac{dv}{dx} + v \frac{du}{dx}$

* Quotient Rule: $\frac{d}{dx} \left(\frac{u(x)}{v(x)} \right) = \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$

First let us look at this: $\frac{d}{dx} \left(\frac{1}{v(x)} \right) = \frac{-dv/dx}{v^2}$

$$\frac{d}{dx} \left(\frac{u(x)}{v(x)} \right) = \frac{1}{v} \frac{du}{dx} - u \frac{dv/dx}{v^2}$$

$$= \frac{v \frac{du}{dx} - u \frac{dv}{dx}}{v^2}$$